Investigating the Biomarker Cascade of Alzheimer's Disease

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Abstract

Alzheimer's disease is a progressive, irreversible brain disorder that gradually destroys cognitive function and memory, making even the simplest tasks difficult to perform. While the general pathological mechanisms underlying Alzheimer's disease are known, the timing and progression of these changes in the brain remain unclear. Previous models suggest that a set of biomarkers evolves over time in a characteristic sequence associated with worsening disease. This framework, known as the biomarker cascade theory, proposes that beta-amyloid and tau aggregation initiate the pathological process, followed by cognitive decline and ultimately death. While experiments have supported this claim, they are limited in their ability to understand the mechanisms underlying the biomarker cascade. Thus, we turn to math modeling to better understand the timing dynamics and interactions of the biomarkers in this cascade. In this study, we analyze a dynamical systems framework to better understand how interactions among biomarkers influence disease progression. By examining the system parameters, we aim to identify which biomarkers exert the greatest influence on the cascade and to explore potential early indicators of Alzheimer's disease, ultimately informing improved treatment strategies.

Keywords: Alzheimer's Disease, Biomarkers, Dynamical Systems Analysis

1. Introduction

Alzheimer's disease (AD) is a major public health concern affecting society, and there is an urgent need to better understand the mechanisms underlying the disease to develop effective therapies [1]. Despite the dozens of clinical drug trials conducted, over the past decade, most have failed, largely due to our limited understanding of the biological processes driving AD [2]. Through genetic studies, researchers identified a general cascade of AD biomarkers known as the "A/T/N" system: "A" denotes β-amyloid biomarker, "T" represents the value of tau biomarker, and "N" encompasses biomarkers of neurodegeneration and cognitive decline [3]. The first two biomarkers can be tracked using PET scans and blood tests by detecting amyloid plaques and tau tangles, as illustrated in Figure 1. Neurodegeneration is typically monitored through MRI and PET scans, while cognitive decline is assessed by various cognitive ability tests. The sequential progression of biomarker changes is described by the dynamic AD biomarker cascade theory, which hypothesizes that AD biomarkers evolve sequentially and in a temporally overlapping manner, typically in the order of A/N/T, and culminate in dementia characterized by full cognitive decline, as illustrated in Figure 2. PET scan studies have refined this model, revealing difference in the accumulation of β-amyloid plaques and tau tangles for Early and Late Onset Alzheimer's. [1] While the "A/N/T" system has been characterized through experiments and clinical trials; there remains a lack in our understanding of how these biomarkers interact and evolve over time.

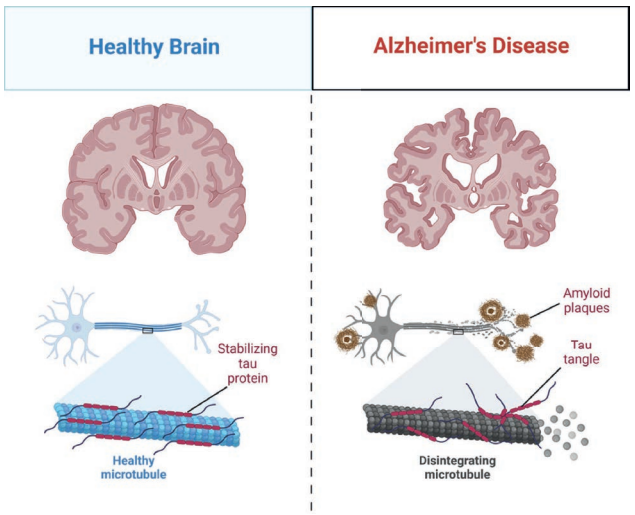


Figure 1: **Alzheimer's Disorder** Healthy brain vs. diseased brain. Includes schematic of amyloid plaques and tau tangles on the neuron [4].

Petrella et al. investigated the temporal dynamics of β-amyloid and tau and their relationship to neurodegeneration and cognitive decline. Since β-amyloid and tau cannot be monitored in the earliest stages of disease, early timing dynamics are difficult to predict, so disease pro-

gression is often monitored through neurodegeneration and cognitive decline. Petrella's group developed a system of five differential equations that capture the growth rates and temporal overlap of each biomarker, using parameters from the Alzheimer's Disease Neuroimaging Initiative. However, their study lacked a steady state analysis and a detailed comparison of the timing dynamics between Early and Late Onset AD.

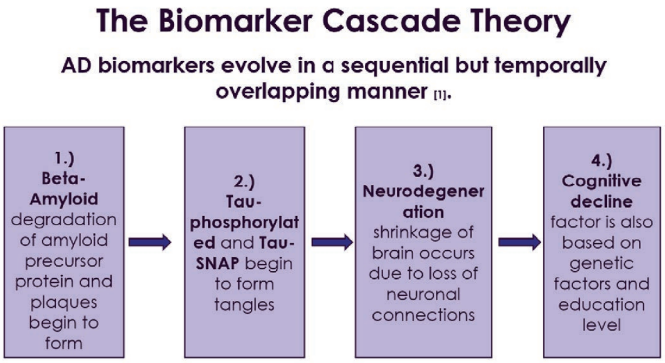


Figure 2: **Biomarker Cascade Theory:** The General cascade of pathological biomarkers of AD. Some forms of Late Onset AD begin with Tau-SNAP (age-related tau tangles) and then β-amyloid plaques forming.

This work aims to extend the analysis of the dynamical system by conducting a steady-state analysis and computing numerical solutions of the model. The steady-state analysis will clarify how growth rates and biomarker interactions affect the stability of the cascade. Additionally, numerical solutions will provide insights into the timing dynamics for both Early-Onset (patients 65 years old and younger) and Late-Onset (patients 65 years and older) AD. Together, these approaches will provide a framework for identifying optimal intervention points for treating and decreasing β-amyloid plaques and tau tangle formation.

2. Early-Onset Alzheimer's Disease Model Analysis

EOAD is classified as a type of AD that affects those under the age of 65 years [2]. EOAD is typically caused by traumatic brain injuries (TBIs) and Alcohol Use Disorder (AUD). These conditions can alter the brain significantly, which leads to changes in brain chemistry and potentially EOAD [2]. EOAD tends to progress slower since age-related effects are not factors; however, the disease still causes irreversible damage that leads to death. EOAD is highly understudied as it affects 5-10 percent of people with AD [2]. Thus, we will be analyzing the cases of the healthy patient(the control case) and the diseased patient. By observing the time-dependent dynamics through a stability analysis and numerical simulations of the healthy and diseased patient, we hope to better understand the biomarker cascade associated with AD when age is not a factor, as is the case in Late-Onset Alzheimer's disease.

2.1. EOAD: The Model, Parameters, and Assumptions

The model for Early On-Set AD is a system of four differential equations adapted from Petrella et al.[1]:

$$\frac{dA_\beta}{dt} = \mu_{A_\beta} A_\beta (K_{A_\beta} - A_\beta) \tag{1}$$

$$\frac{dT}{dt} = \lambda_{TA_\beta} A_\beta + \mu_T T(K_T - T) \tag{2}$$

$$\frac{dN}{dt} = \lambda_{NT} T + \mu_N N(K_N - N) \tag{3}$$

$$\frac{dC}{dt} = \lambda_{CN} NR + \mu_C C(K_C - C) \tag{4}$$

where K_* denotes the carrying capacity, μ_* represents the individual growth rate of each biomarkers, λ_* quantifies the interactions between biomarkers, and R is the cognitive reserve term, which depends on a patient's genetics and educational background. Each equation in (1)-(4) follows a logistic growth model and the cascade is enforced though the temporally overlapping interaction terms λ_*. Figure 3 provides a schematic of the cascade and how each parameter fits into the model.

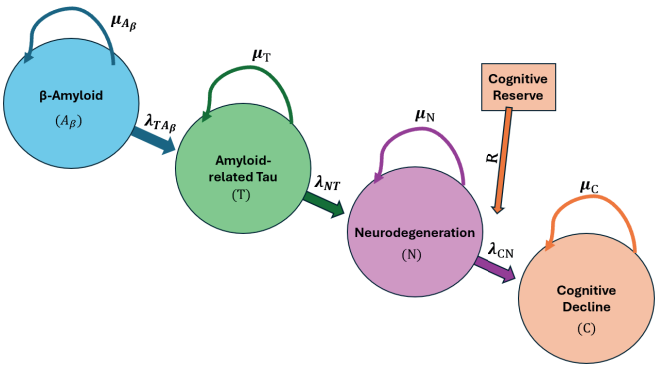


Figure 3: **EOAD Model Schematic:** Schematic of the four-dimensional model for EOAD.

The assumptions underlying this model include: (1) the growth of the biomarker cascade is implemented through a logistic growth model of carrying capacity K_* for each component, (2) The minimal biomarker level on graphs is set to 0.05 for β-amyloid to allow for various onset delays, (3) Cognitive reserve, R, is a constant that modifies the effect of neuronal degeneration on the onset of cognitive decline, with lower values indicating lower risk and higher values indicating higher risk group. Model parameters, summarized in Table 1 , are derived from PET scan data from the Alzheimer's Disease Neuroimaging Institute (ADNI).

Parameters	Description	Values
$K_{A_\beta,T,C,N}$	carrying capacities for general parameters	1
μ_{A_β}	amyloid pathological cascade	0.08
μ_T	tau (amyloid-related) pathological cascade	0.05
μ_N	neurodegeneration cascade	0.05
μ_C	cognitive impairment cascade, growth rate	0.05
λ_{TA_β}	tau from amyloid, growth rate	0.025
λ_{NT}	neurodegeneration from amyloid-related tau	0.025
λ_{CN}	cognitive impairment from neurodegeneration	0.001
R_1	cognitive reserve (low risk)	1
R_2	cognitive reserve (high risk)	25

Table 1: Model parameters for Early-Onset Alzheimer's Disease taken from studies done by the Alzheimer's Disease Neuroimaging Initiative (ADNI) [1, 5].

The strong cascade is represented through the λ_* terms and the logistic growth terms with carrying capacity K_*. Notably, setting $A_\beta = 0$ in Equation (2), results in Tau (T) reaching the carrying capacity if the initial value of tau is greater than 0; this pattern extends to Equations (3) and (4) as well. Biologically, this tells us that disease biomarkers will always reach their carrying capacity if there is an initial value of A_β or a small initial value of each biomarker-an outcome less relevant biologically.

A stability analysis of the model will reveal which parameters and biomarkers influence the system stability or determine whether a patient on a threshold of disease will progress further. This is essential for understanding the timing of when tau, neurodegeneration, and cognitive decline reach their carrying capacities. The analysis will also indicate which parameters may be accelerating disease and inform us of potential intervention points for treatment; thus, informing where treatment dynamics should be implemented.

In the following sections of the EOAD model, a stability analysis will be performed to determine if there are any parameters that destabilize the system in patients who already have the disease. Stability biologically represents the continued progression of disease. If we can uncover parameters that affect stability when the disease is already in progress, this will inform us of the treatment methods and the overall inner workings of the disease that are not yet accessible experimentally.

2.2. Patient Cases

2.2.1 The Healthy Patient

To understand the effect of the biomarker cascade on a diseased patient, we first analyze the case of the healthy patient. We can look at this case first as our control, meaning that none of the biomarker cascades have reached their carrying capacity, which is represented by the equilibrium point $(0, 0, 0, 0)$. Thus, to analyze the stability of the healthy patient case, we find the general Jacobian to be

$$\mathrm{J}(\mathrm{A}_\beta, T, N, C) = \begin{bmatrix} \mu_{A_\beta} K_{A_\beta} - 2\mu_{A_\beta} A_\beta & 0 & 0 & 0 \\ \lambda_{TA_\beta} & K_T\mu_T - 2\mu_T T & 0 & 0 \\ 0 & \lambda_{NT} & K_N\mu_N - 2\mu_N N & 0 \\ 0 & 0 & \lambda_{NC} R & K_C\mu_C - 2\mu_C C \end{bmatrix} \tag{5}$$

then plug in the equilibrium point $(0, 0, 0, 0)$, and find the corresponding eigenvalues to determine the stability. Knowing that the eigenvalues of a triangular matrix (counting multiplicities) are exactly the diagonal entries [6], we find the eigenvalues to be

$$\begin{aligned} \lambda_1 &= \mu_{A_\beta} K_{A_\beta} \\ \lambda_2 &= K_T\mu_T \\ \lambda_3 &= K_N\mu_N \\ \lambda_4 &= K_C\mu_C. \end{aligned}$$

We can now use the real parts of the eigenvalues to assess the stability of a healthy patient. Since all parameters are positive, all the eigenvalues obtained are positive as well. Since we have four positive eigenvalues, this informs us that $(0, 0, 0, 0)$ is an unstable point as all trajectories move away from that point. In Figure 4(a)-(c), we can observe this effect as any small perturbation of A_β from 0 will cause the other biomarkers to also reach some stable point. Biologically, this means that any amount of β-amyloid plaques that begin to form will lead to the cascade. There are other non-biologically relevant stable points where, if we perturb the other biomarkers, they will lead to different stable points for the biomarkers that follow the perturbed one in a cascade fashion, always leaving prior biomarkers in the cascade at 0. However, these are not biologically relevant since A_β always starts the cascade. These other stable points are briefly discussed in **Section** 2.2.3.

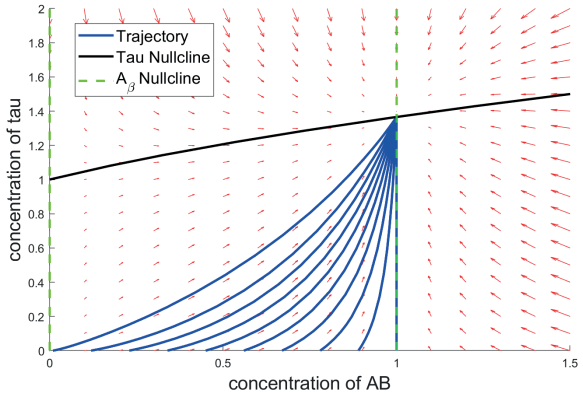
2.2.2 Patient with Fully Progressed Disease

The next case we will analyze is the diseased patient. In the mathematical context of the model, this is when the β-amyloid reaches its carrying capacity, $K_{A\beta}$. Thus, using a similar process as above, we find the equilibrium point for the fully diseased patient to be

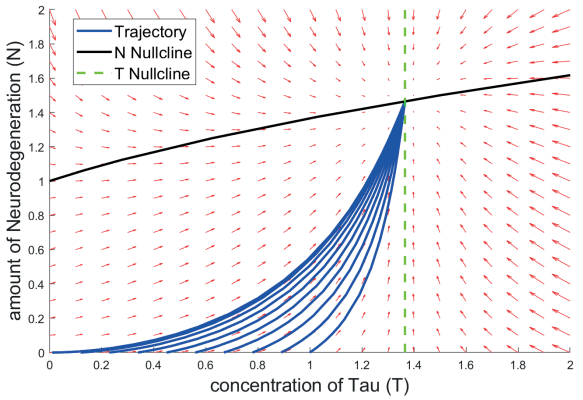
$$(K_{A_\beta}, T^*, N^*, \frac{K_C(1 + \sqrt{1 + 4\delta_3 N^*})}{2}),$$

where

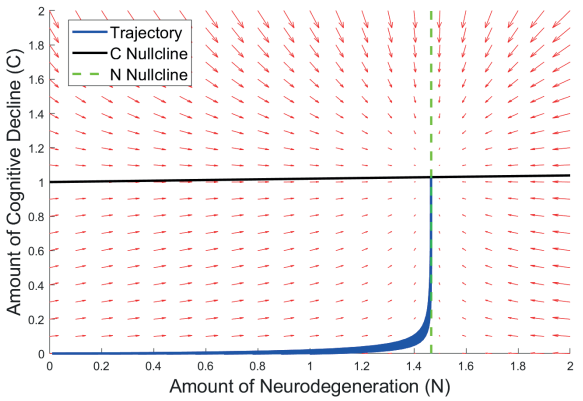
$$\begin{aligned} T^* &= \frac{K_T(1 + \sqrt{1 + 4\delta_1})}{2} \\ N^* &= \frac{K_N(1 + \sqrt{1 + 4\delta_2 T^*})}{2} \\ \delta_1 &= \frac{\lambda_{TA_\beta} K_{A_\beta}}{\mu_T K_T^2} \\ \delta_2 &= \frac{\lambda_{NT}}{\mu_N K_N^2} \\ \delta_3 &= \frac{\lambda_{CN} R}{\mu_C K_C^2}. \end{aligned}$$



(a)



(b)



(c)

Figure 4: **Phase Portraits to Observe Stability:** the black lines represent the nullclines for tau in (a), neurodegeneration (b), and cognitive decline (c) and the green represent the equilibrium point when $A_\beta = K_{A_\beta}$ for the other biomarker graphed in a, b, and c.

Now that we have obtained the equilibrium points for the fully diseased patient case, we can use Jacobian to find the eigenvalues, which will inform us of the stability in this case.

The eigenvalues obtained from the Jacobian (see full paper link for more details on the mathematical process) are:

$$\lambda_1 = \mu_{A_\beta} K_{A_\beta} - 2\mu_{A_\beta} K_{A_\beta}$$
$$\lambda_2 = K_T \mu_T - 2\mu_T T^*$$
$$\lambda_3 = K_N \mu_N - 2\mu_N N^*$$
$$\lambda_4 = K_C \mu_C - 2\mu_C \left(\frac{K_C(1+\sqrt{1+4\delta_3 N^*})}{2}\right).$$

Since μ_*, λ_*, R, and K_* are always positive values, all eigenvalues equate to

$$\mu_*(K_* - K_*(1+\sqrt{>0})) = -K_*\mu_*(\sqrt{>0})). \tag{6}$$

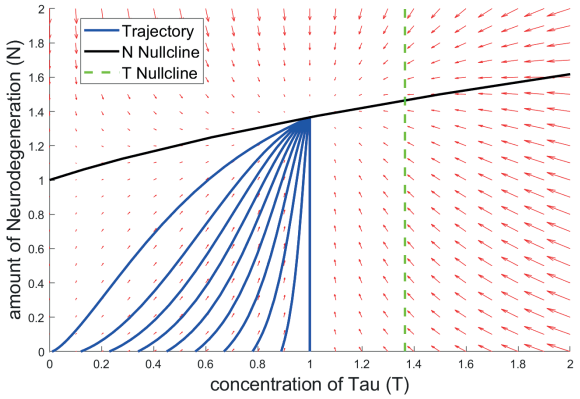
Thus, all eigenvalues will be negative for any parameters. This result informs us that when β-amyloid reaches its carrying capacity, K_{A_β}, each subsequent biomarker will reach some stable state as defined by the equilibrium point. Additionally, since the eigenvalues are negative for any parameters, the system is always stable at this fixed point. This result begins to answer our question about the stability of disease, meaning that once the cascade has begun, β-amyloid has a value above 0. Then every biomarker will reach a stable point where disease will neither progress further nor decrease. Stability is shown further in the phase diagrams in Figure 4 (a)-(c), as soon as there is a small amount of β-amyloid, then β-amyloid will reach its carrying capacity K_{A_β} as its stable point. The subsequent points for tau, neurodegeneration, and cognitive decline also end up reaching the points found from the equilibrium point.

Thus, we can conclude that once β-amyloid has reached it carrying capacity, then all biomarkers will reach a stable point where disease is neither continuing to progress nor decreasing. In the following section, we will observe the other cases where each biomarker has reached its carrying capacity; however, these cases are not biologically relevant as β-amyloid always starts the cascade. These cases will further prove the concept of the strong cascade that occurs, which is important for developing treatment methods.

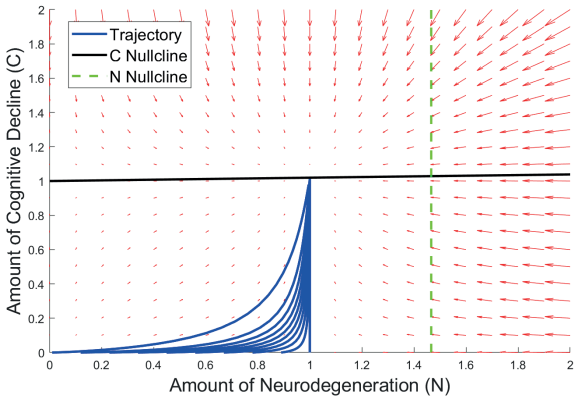
2.2.3 Extraneous Cases

There are three other cases that are not relevant biologically, but can be explored mathematically. The three cases occur when β-amyloid does not start the cascade. The biomarkers following β have three different fixed points where each is set to their carrying capacity, K_*. Essentially, if we were to perturb from the healthy patient by increasing one of the biomarkers, then that one will go to its carrying capacity and subsequent biomarkers will reach a different fixed point than found in the diseased patient example. Thus, if the perturbation occurs early in the cascade, at Tau for instance, then it will force the

other biomarkers to reach at least their carrying capacities. However, if only cognitive decline is perturbed, all other biomarkers will remain at 0.



(a)



(b)

Figure 5: **Trajectories of The Extraneous Cases:** for the equilibrium points where $T = K_T$ and $N = K_N$ compared to the equilibrium points in Figure 4. The green and black nullclines represent the trajectories for T, N, and C in the case of $A_\beta = K_{A_\beta}$. (a) represents when $T = K_T$ and the trajectories end up at a different stable point for N and T as compared to those in Figure 4, represented by the green and black nullclines. (b) represents a similar comparison, except $N = K_N$ and $A_\beta = T = 0$.

Essentially, any small perturbation from one of the biomarkers will lead to subsequent biomarkers reaching at least their carrying capacity, as seen in Figure 5 (a)-(b). This is an important aspect of the model when thinking about treatment terms that could affect stability of the system. If there is a small amount of β-amyloid removed by treatment or even none left in the system, and if the other biomarkers have already been perturbed, then they will still reach their carrying capacities. This begins to indicate to us that timing dynamics will be critical to observe when it comes to treating EOAD as once the disease begins, the biomarkers will always reach at least carrying capacities.

2.3. Time Dynamics of Diseased Patient with EOAD

Numerical solutions provide us with a way to visualize the biomarker cascade of AD over time. To find the numerical solution, we use the parameters from Table 1. Figure 6 provides a simulation for the EOAD case where all values go to 1. We normalized the curves by dividing the max value, which is the stable point reached for each biomarker. In any case of AD, effects of dementia, reduced memory loss and decreased cognitive ability, usually occur when cognitive decline is at 0.6. We can see that for those with EOAD and a lower R-value, representing higher educational level and other cognitive factors, symptoms of dementia do not start until age 100. Those with higher R-values, however, will experience cognitive decline effects between 50 and 80 years of age. For all values of R, we can see that neurodegeneration, which leads to death due to shrinkage of brain, starts to be detected around 45 years and rapidly increases. The biomarker cascade and logistical growth of these biomarkers also stay prevalent as we can see rapid growth occurring and stability at their stable points before the age of 65.

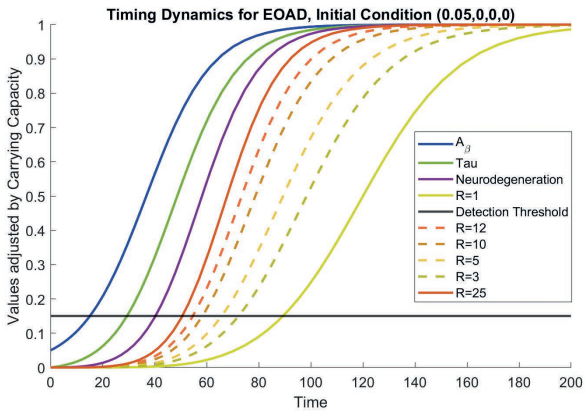


Figure 6: **Numerical Solution for EOAD Model:** solved using ODE45 solver in MATLAB using parameters in Table 1, where time is relative to the start of the disease.

In Figure 7, we can see the strength of the cascade, as even if we do not start with β-amyloid, but instead we have an initial value of Tau, the cascade will follow, allowing all the biomarkers to reach a different stable point than the one in Figure 5. Essentially, these solutions support the idea that when the cascade begins or if tau or other biomarkers are pushed in some way, they will always end up reaching at least their carrying capacities. Thus, now that we have the general timing dynamics of

the EOAD model, we can compare them to those of the LOAD model in the following sections. Additionally, as we

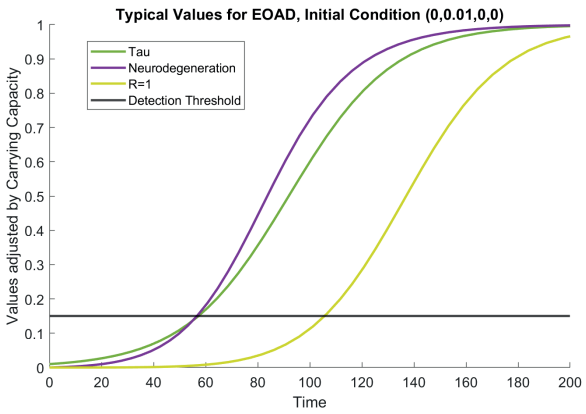


Figure 7: **Numerical solution for EOAD Model with $A_{\beta_0} = 0$:** solved using ODE45 solver in MATLAB using parameters in Table 1 with time in years. This represents the solution when Tau has an initial value, but β-amyloid does not.

found in the diseased patient section, parameters cannot affect the stability, but they could affect the timing dynamics if a treatment term were to be added to decrease growth rate of β-amyloid for instance. Thus, as we move to the next sections, it is important to keep in mind that once the cascade has begun all values will reach their stable point and cannot be perturbed from this point.

3. Late-Onset Alzheimer's Disease Model Analysis

LOAD affects people aged 65 years and older [2]. Around 80 percent of cases of AD occur within this age group; thus, most of the data from the ADNI comes from patients in this group. Patients with LOAD typically develop the disease due to a combination of genetic, environmental, and lifestyle factors. The primary genetic risk factor for LOAD is the apolipoprotein E ($APOE$) $\epsilon 4$ allele [7]. This genetic biomarker has been studied and is included in Petrella et al. (2019)'s model; however, in our work, we ignore this factor, as many cases of LOAD are not associated with this genetic factor.

LOAD primarily differs from EOAD as there is a distinct type of tau affecting disease progression, Tau-SNAP, also known as Aging SNAP [1]. Tau-SNAP is a form of neurofibrillary tangles that are not associated with AD but occur naturally with age. In this model, we incorporate another ODE that represents total tau, including Tau-SNAP and phosphorylated tau associated with AD. While the model follows the same general biomarker cascade, Tau-SNAP also affects neurodegeneration and, consequently, cognitive decline. Although Tau-SNAP does not necessarily initiate the cascade, most patients typically have some level of Tau-SNAP at baseline, even if they do not have β-amyloid plaques initially.

Now that we have incorporated age-related factors into the model, we perform another stability analysis. Timing dynamics of LOAD progression will also be compared to those of EOAD. By examining age-related tau factors, we aim to determine whether Tau-SNAP influences the stability of the steady state, which was not observed in the EOAD case. Thus, by analyzing differences in equilibrium points and steady states, we seek to gain a better understanding of the cascade, the effects of Tau-SNAP, and when to incorporate treatment.

3.1. LOAD Model, Parameters, and Assumptions

The model for the LOAD patient is a system of five differential equations:

$$\begin{aligned}
\frac{dA_\beta}{dt} &= \mu_{A_\beta} A_\beta (K_{A_\beta} - A_\beta) \\
\frac{dT}{dt} &= \lambda_{TA_\beta} A_\beta + \mu_T T(K_T - T) \\
\frac{dT_{all}}{dt} &= \lambda_{TA_\beta} A_\beta + \mu_T T_{all}(K_T - T_{all}) + \lambda_{T_o AS} AS \\
\frac{dN}{dt} &= \lambda_{NT_o}(T_{all} - T) + \lambda_{NT} T + \mu_N N(K_N - N) \\
\frac{dC}{dt} &= \lambda_{CN} NR + \mu_C C(K_C - C)
\end{aligned}$$

where λ_* incorporates the rate of Tau-SNAP corresponding to AS, which is Aging Snap and neurodegeneration, λ_{NT_o}. As seen in Figure 8, we can see the model generally operates in a similar manner to the EOAD as the LOAD model consists of logistic growth and a linear term that represents the connection between each biomarker. The only difference is this additional age-related tau that is already in the system and grows linearly rather than logistically. Thus, this term incorporates additional assumptions including: (1) the starting value of β-amyloid is 0.01 and 0.05 for Tau-SNAP, (2) Tau-SNAP affects neurodegeneration and, consequently, cognitive decline. In addition to these assumptions, we utilize the parameters in Table 2, also taken from the ADNI.

In the following sections, we perform a stability analysis of the LOAD model to better understand the role Tau-SNAP plays in the cascade. If Tau-SNAP plays a role in changing the stability of the cascade, this could inform us that LOAD has different dynamics than that of EOAD, which always reaches a stable point. This is important as we can gain further insight into the dynamics of LOAD and where potential treatment could be applied. Additionally, if Tau-SNAP itself does not affect the cascade, we can observe how parameters affect the stability of the

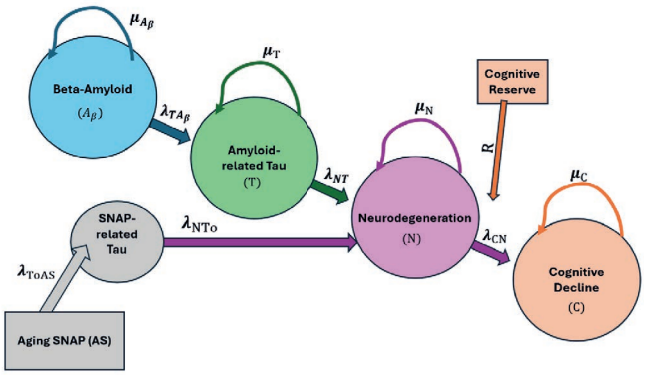


Figure 8: **LOAD Model Schematic:** Schematic of the five-dimensional LOAD Model

system, which could also inform of where to include additional treatment terms that could affect the stability of the system. Along with the stability analysis, we will then later analyze the timing dynamics of LOAD as compared to EOAD. Based on the parameters and the additional tau term, we hypothesize that LOAD progresses faster than EOAD. Thus, by understanding the stability of the system with an extra tau term and the timing dynamics as compared to EOAD, we will understand the role that age-related effects play on disease progression.

Parameters	Description	Values
$K_{A_\beta,T,C,N}$	carrying capacities for general parameters	1
T_{o0}	Tau(SNAP-related) pathology, initial value	0.05
μ_{A_β}	amyloid pathological cascade	0.12
μ_T	tau(amyloid-related)pathological cascade	0.05
μ_N	Neurodegeneration cascade	0.05
μ_C	Cognitive impairment cascade, growth rate	0.05
λ_{T_oAS}	Tau (SNAP-related) path from aging/SNAP, growth rate	0.002
λ_{NT_o}	Neurodegeneration from tau (SNAP-related), growth rate	0.0075
λ_{TA_β}	tau (amyloid-related) from amyloid, growth rate	0.025
λ_{NT}	Neurodegeneration from amyloid-related tau	0.025
λ_{CN}	Cognitive impairment form neurodegeneration	0.001
AS	Aging SNAP	1
R_1	Cognitive reserve (low risk)	1
R_2	Cognitive reserve (high risk)	25

Table 2: Model parameters for Late-Onset Alzheimer's Disease taken from studies done by the Alzheimer's Disease Neuroimaging Initiative (ADNI)][1, 5].

3.2. Patient Cases

3.2.1 The Healthy Patient

Similarly to the EOAD stability analysis, first we will observe the healthy patient case, which is (0,0,0,0). In the EOAD case, this point was unstable, meaning that any small perturbation to β-amyloid will cause each biomarker to go to a stable point. Following a similar stability analysis as performed before, we find the general Jacobian of the system to be

$$J(A_\beta,T,N,C) = \begin{bmatrix} \mu_{A_\beta}K_{A_\beta} - 2\mu_{A_\beta}A_\beta & 0 & 0 & 0 & 0 \\ \lambda_{TA_\beta} & K_T\mu_T - 2\mu_T T & 0 & 0 & 0 \\ \lambda_{TA_\beta} & K_T\mu_T - 2\mu_T T & \mu_T K_T - 2\mu_T T_{all} & 0 & 0 \\ 0 & \lambda_{NT} - \lambda_{NT_o} & \lambda_{NT_o} & K_N\mu_N - 2\mu_N N & 0 \\ 0 & 0 & 0 & \lambda_{NC}R & K_C\mu_C - 2\mu_C C \end{bmatrix}. \tag{7}$$

Once we plug in (0,0,0,0) into the Jacobian and plug it into $det(J - \lambda I) = 0$, we find the eigenvalues of the Healthy patient to be

$$\begin{aligned} \lambda_1 &= \mu_{A_\beta}K_{A_\beta} \\ \lambda_2 &= \lambda_3 = K_T\mu_T \\ \lambda_4 &= K_N\mu_N \\ \lambda_5 &= K_C\mu_C. \end{aligned}$$

Since all eigenvalues are positive, this implies that the equilibrium point $(0,0,0,0,0)$ is unstable, like the EOAD. Biologically, this means that a small perturbation to β-amyloid will cause the other biomarkers to also begin to cascade. This is an interesting fact as many people during their lifetime will develop β-amyloid plaques that could set off this cascade as they grow older, but not everyone develops Alzheimer's disease. Thus, this informs us that timing dynamics of this push from the unstable state will be important for understanding LOAD progression. Another interesting result is that of $lambda_2$ and λ_3 being equivalent to one another. However, this is to be expected as Tau_{all} encapsulates both types of taus, related and non-related to Alzheimer's Disease. Thus, since this term integrates the damage of Tau-SNAP through a linear term that is not affected by the β-amyloid, then only the Tau related to AD will have a role in affecting the stability of the healthy patient. Meaning, if tau related to AD were to be perturbed, then this could cause neurodegeneration and cognitive decline to reach a stable point, but the same is not true for the Tau-SNAP. While Tau-SNAP does affect neurodegeneration and consequently cognitive decline, the rate times at which this grow is significantly slower than that of Tau related to β-amyloid and AD, as seen in Figure 9.

3.2.2 The Diseased Patient

The next case is the diseased patient. This occurs when we have the equilibrium point where A_β is equivalent to the carrying capacity, K_{A_β}. In this case, we find the equilibrium point to be

$$(K_{A_\beta}, T^*, T^*_{all}, N^*, C^*)$$

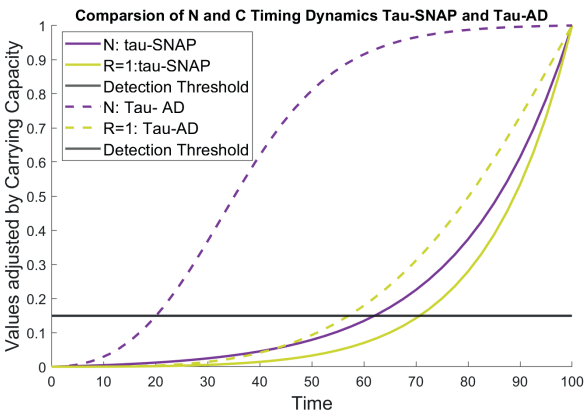


Figure 9: **Comparison of Tau-SNAP and Tau Related to AD :** Comparison of timing dynamics to show the slower progression of neurodegeneration when Tau-SNAP is the only factor compared to the dotted lines which is when tau related to AD is the only type present.

where each value is equal to,

$$T^* = \frac{K_T(1+\sqrt{1+4\delta_1})}{2}$$

$$T^*_{all} = \frac{K_T + \sqrt{K_T^2 + 4(\frac{1}{\mu_T}(\lambda_{TA_\beta}K_{A_\beta} + \lambda_{T_oAS}AS))}}{2}$$

$$N^* = \frac{K_N + \sqrt{K_N^2 + 4(\frac{1}{\mu_N}(\lambda_{NT_o}(T^*_{all} - T^*) + \lambda_{NT}T^*)}}{2}$$

$$C^* = \frac{K_C\sqrt{K_C^2 + 4\frac{\lambda_{CN}R}{\mu_C}N^*}}{2}$$

$$\delta_1 = \frac{\lambda_{TA_\beta}K_{A_\beta}}{\mu_T K_T^2}.$$

Analytically, we can determine the stability of the system at this point using the same process as the Healthy Patient and Equation (7). After plugging in the equilibrium point for when $A_\beta = K_{A_\beta}$ we obtain the following eigenvalues:

$$\lambda_1 = \mu_{A_\beta}K_{A_\beta} - 2\mu_{A_\beta}K_{A_\beta}$$
$$\lambda_2 = K_T\mu_T - 2\mu_T T^*$$
$$\lambda_3 = K_T\mu_T - 2\mu_T T^*_{all}$$
$$\lambda_4 = K_N\mu_N - 2\mu_N N^*$$
$$\lambda_5 = K_C\mu_C - 2\mu_C C^*$$

Like the EOAD case, each eigenvalue equates to Equation (6) which indicates that all eigenvalues are negative. Thus, similarly to the EOAD case, we can conclude that the equilibrium points for which $A_\beta = K_{A_\beta}$ is a stable point. In the biological context, this means that once the cascade begins from perturbation of β-amyloid, then all biomarkers will reach some steady state above their carrying capacities. Phase portraits that confirm this result are in the link to the full paper. Additionally, like the case for EOAD, there are no parameters that can affect stability. One though we had was that Tau-SNAP may affect the stability at this fixed point; however, based on the eigenvalues, there are no parameters correlated to Tau-SNAP (λ_{T_oAS}, $lambda_{NT_o}$, and AS) that could affect stability. All these parameters are under the square root, meaning that these values will never change the sign of the eigenvalue. The same is true for all other parameters, indicating that EOAD and LOAD follow the same behavior where a perturbation to β-amyloid will cause the whole cascade to reach a stable point, as defined by the equilibrium point for the diseased patient.

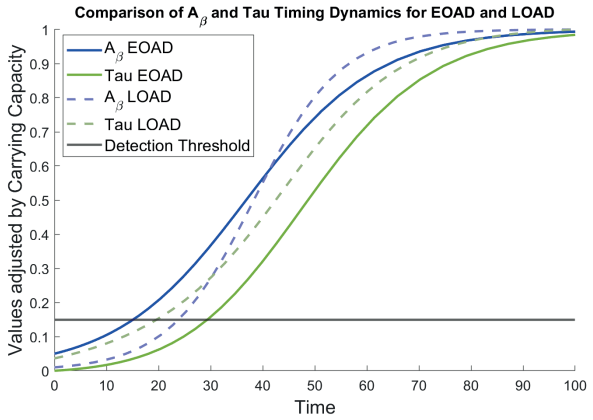
3.3. Timing Dynamics of Late On-Set AD Compared to Early On-Set AD

Since the stability of the model has been determined, we can now analyze the numerical solution to understand the timing dynamics behind of the LOAD cascade. In Figure 10 (a) and (b), we observe a faster rate at which all biomarkers progress, as the dashed lines represent the LOAD model and parameters from Table 2. We observe faster dynamics due to the β-amyloid growth rate being higher for those with LOAD and due to the extra Tau-SNAP term. For the β-amyloid, it is generally known that all older adults will eventually develop some level of β-amyloid. Thus, depending on the concentration of β-amyloid and the age at which these plaques form can affect the timing dynamics. However, for all individuals with LOAD β-amyloid will reach its carrying capacity at a much faster rate than EOAD since it is a factor affected by age.

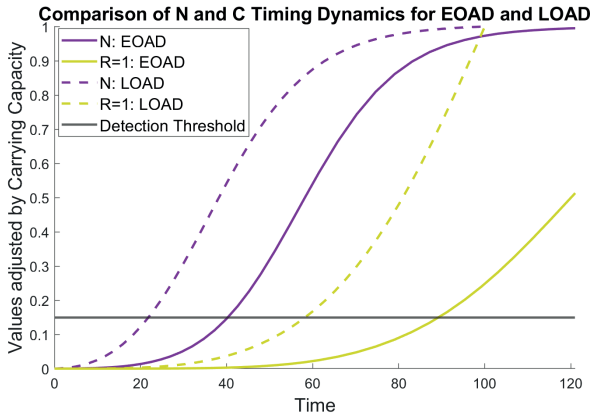
The tau-SNAP, also an age-related factor, already adds in a base value of Tau to the system. Thus, even though the growth rate of tau is the same for both EOAD and LOAD, since there is initial tau in the system, we see a drastic increase. At points, since the initial Tau-SNAP is larger than that of A_β, we see interesting behavior where it seems that Tau-SNAP increases faster than β-amyloid, unlike in the EOAD case that follows the usual cascade. However as both β-amyloid and tau total increase, β-amyloid ends up increasing faster than Tau. This is again since the Tau-SNAP is also a constant term.

A similar trend is seen between neurodegeneration and cognitive decline. Since neurodegeneration is affected by Tau-SNAP, this increases the rate at which brain shrinkage occurs, and consequently the rate at which cognitive decline occurs. The rate of cognitive decline is much faster than that of EOAD and even reaches the value 0.6, which symbolizes Dementia-like symptoms significantly earlier than the EOAD model.

Overall, the timing dynamics of LOAD are faster than those of EOAD due to age-related factors of tau and β-amyloid. Additionally, this is important in knowing when



(a)



(b)

Figure 10: **Comparison of Timing Dynamics between LOAD and EOAD for all Biomarkers:** The dotted line represents the LOAD with initial condition (0.01,0,0.05, 0, 0) and parameters from Table 2, while the solid lines represent EOAD with the initial condition (0.05,0,0,0) and parameters from Table 2. (a) Compares A_β and tau timing dynamics and (b) compares neurodegeneration and cognitive decline timing dynamics. All solutions were plotted in MATLAB using the ODE45 solver.

and where to add treatment for both factors based on the speed at which the biomarkers progress. While both EOAD and LOAD can never be pushed away from the stable state, once the cascade has begun, there may be a way to affect parameters that slow down timing dynamics of each biomarker.

4. Discussion

Overall, we sought out to understand the timing dynamics and stability of the biomarker cascade associated with Alzheimer's Disease. With tau, there remains unknowns about the link between amyloid and tau as the precise mechanism that triggers tau pathology remains unclear [8]. Additionally, there remains unknowns as to why some patients develop β-amyloid plaques but never reach the level of cognitive decline where dementia is reached [8]. Thus, we will explore how the analysis performed in this paper connects to these ideas and informs us of future treatment methods.

4.1. Stability of the Cascade

Through a stability analysis of both EOAD and LOAD models, we successfully gained a better understanding of how biomarkers interact with each other. Biological reasons as to why β-amyloid pushes tau to cascade were unknown from this model, but through the stability analysis, we observed that the cascade between β-amyloid and tau is inevitable. This inevitable cascade is based on trajectories from the phase portraits in Figure 4, where we concluded that as soon as β-amyloid is perturbed, the system will reach some equilibrium point. However, our analysis lacks an understanding of what that exact concentration or amount of β-amyloid causes for the push of the other biomarkers, which is why biological reasons as to why β-amyloid pushes tau remains unknown.

4.2. Time Dynamics of Disease

From the stability analysis, we observed the importance of understanding the timing dynamics of disease. As mentioned, there is a gap in our understanding of why some patients develop cognitive decline and others do not. Based on Figures 6 and 10, we can see that timing is key for when cognitive decline comes into play as well as the R-value. The R-value, as seen in Figure 6, plays a key role in when symptoms of dementia begin to be expressed. For instance, at R=25 cognitive decline can start relatively 60 years from the beginning of the β-amyloid plaque formations and up to 120 years later for R = 1. Thus, being able to classify one's level of cognitive reserve is important for understanding the time dynamics behind why some may develop dementia quicker than others. Additionally, these timing dynamics, when comparing EOAD and LOAD informed us that with faster growth rates of β-amyloid and tau, this can significantly increase the rate at which neurodegeneration and cognitive decline occur. For instance, in Figure 10 (a) and (b), we observed that even though the β-amyloid and tau reached their stable point quicker in LOAD as compared to EOAD, the neurodegeneration and cognitive decline were significantly faster. This informs us that possibly finding a way to decrease the growth rates of β-amyloid and tau in LOAD could slow down the dynamics of the neurodegeneration and cognitive decline. Overall, through analysis of the timing dynamics between LOAD and EOAD provided insights into future work on adding a treatment term to β-amyloid and tau for LOAD.

Findings about the addition of a treatment term to these two biomarkers can be found in the full paper below.

4.3. Future Direction: A Treatment Term

Based on both the analyses mentioned in this paper on EOAD and LOAD, in the full paper I observe the timing dynamics when a treatment term is applied to the β-amyloid and tau terms. Treatment options for those with LOAD are offered, however, they are only given to those who have low to no cognitive decline [9]. These treatments attack the β-amyloid plaques and have been shown to help slow cognitive impairment. In addition to these treatments, that have been approved by the FDA, clinical trials are currently happening for a drug that combines both suppresses for tau and β-amyloid [10]. While data is not available yet for this specific study, we hope by adding a Heaviside and exponential treatment term to both tau and β-amyloid, we can observe how timing dynamics change when both biomarkers are affected compared to only β-amyloid.

Access Full Paper

https://drive.google.com/drive/folders/17oiUYPCObQy2tNKC85T71m2QMPHHylCk?usp=drive_link

KRYSTA PEREIRA



My first memory of math occurred in first grade when everyone flipped their behavior sticks from green to yellow because a substitute teacher said we were all in trouble. My stick had never turned yellow in my first two years of elementary school, so I was devastated. That changed when the substitute brought out addition and subtraction flashcards and made a deal: beat your partner, and you could turn your stick back to green. I perked up and, by the end, my stick turned green. In that moment, I felt confident and excited by the challenge. It was the first time I remember feeling exhilarated by math. From then on, I looked for ways to stretch my brain in math. When I first took Euclidean and Non-Euclidean Geometries at Middlebury, I initially felt confident but soon fell into imposter syndrome. At home, I had been used to advocating for myself, but here everything felt different. I didn't know how to navigate asking for help or approach peers. I felt overwhelmed and as though my love for math was fading. After that semester, I began to rediscover my love for math in a new way, by finding a community of people who shared my joy in mathematics. My peers' encouragement helped me reconnect with my passion and motivated me to continue pursuing math after graduation, transforming moments where I almost gave up into renewed joy and love for the subject. Without the support of my mentors, peers, and family, I would have likely stopped pursuing mathematics much earlier. I have learned the importance of community, passion, and perseverance, and I am excited to continue pursuing math and finding communities like the one I found at Middlebury.